Luigui **Gallardo-Becerra**

Molecular Biologist | Bioinformatician

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Molecular Biologist and Bioinformatician with 6+ years of experience integrating wet-lab techniques with advanced computational analysis. Skilled in experimental design, NGS library preparation, PCR/qPCR, microbial culture, and molecular cloning, combined with strong expertise in genomics, transcriptomics, and microbiome analysis. Experienced in building reproducible bioinformatics workflows and analyzing large-scale biological datasets to generate actionable scientific insights. Author of 14 peer-reviewed publications and contributor to interdisciplinary projects across molecular biology, microbiology, and computational biology.

Research Experience

Research Assistant (Molecular Biology & Bioinformatics) — *Institute of Biotechnology, UNAM*

January 2019 - July 2025

Bioinformatics workflow support: Designed and maintained reproducible data-processing workflows using Snakemake, Nextflow, and Bash to support molecular biology and genomics projects.

NGS data interpretation: Analyzed next-generation sequencing datasets—including RNA-seq, whole genome sequencing, 16S rRNA profiling, metagenomics, and metatranscriptomics—to extract biologically meaningful insights that informed experimental design and downstream validation.

Microbiome and functional analysis: Performed taxonomic profiling, functional annotation, and comparative metagenomics to characterize microbial communities and identify gene functions relevant to host–microbe interactions.

Statistical and machine learning applications: Applied statistical modeling and supervised/unsupervised learning approaches to identify biomarkers, classify phenotypes, and evaluate molecular signatures.

Data and database management: Built and maintained curated biological datasets and internal databases, ensuring high data quality, traceability, and reproducibility across projects.

High-performance computing operations: Managed HPC environments used for large-scale sequence analysis, including job scheduling, software installation, workflow optimization, and troubleshooting.

Custom computational tools: Developed targeted analysis scripts and utilities in Python, R, Bash, and Perl to streamline laboratory data processing, quality control, and reporting.

Data Engineer — *Appen*

October 2016 - December 2018

Supported large-scale data processing projects by developing automated ETL pipelines to ensure efficient integration, cleaning, and transformation of high-volume datasets.

Implemented data quality control workflows used for machine-learning model development, improving reliability and reproducibility of downstream analyses.

Designed automated data collection and validation systems that reduced manual workload, increased data accuracy, and streamlined reporting processes across projects.

Software Engineer (Internship) — *Dept. of Computer Science, CUCEI*

January 2016 - July 2016

Developed a cloud-based web application using ASP.NET Core for managing patient records and associated clinical data, contributing to database design, backend development, and user interface implementation.

Built features to support secure data storage, retrieval, and visualization, reinforcing best practices in data integrity and compliance—skills directly applicable to modern bioinformatics and laboratory information systems.

Education

National Autonomous University of Mexico (UNAM) *Mexico City, Mexico*
PH.D. IN COMPUTATIONAL BIOLOGY *January 2019 - July 2025*

National Autonomous University of Mexico (UNAM) *Mexico City, Mexico*
MASTER OF SCIENCE IN COMPUTATIONAL BIOLOGY *August 2016 - January 2019*

University of Guadalajara (UDG) *Guadalajara, Mexico*
BACHELOR OF SCIENCE IN MOLECULAR BIOLOGY *August 2012 - January 2016*

Key Skills

Molecular Biology Techniques:
DNA/RNA extraction, PCR/qPCR, gel electrophoresis, NGS library prep, sample QC, bacterial culture, plasmid prep, sterile technique (BSL-2), experimental design, troubleshooting.

NGS & Genomics Analysis:
QC (FastQC, MultiQC), read alignment (BWA, Bowtie2, HISAT2), variant calling, de novo assembly (Trinity, SPAdes), RNA-seq, 16S profiling, metagenomics, metatranscriptomics, viromics.

Bioinformatics & Workflow Development:
Snakemake, Nextflow, QIIME2, Bioconductor, Biopython, samtools, BLAST, Kraken2, reproducible pipelines, ETL automation.

Programming & Scripting:
Python, R, Bash, SQL, C#, Java, JavaScript, HTML/CSS.

Computing & Infrastructure:
Linux/Unix, HPC (SLURM), Docker, Conda, Git/GitHub, CI/CD, AWS, Google Cloud.

Data Analysis & Visualization:
Statistics, machine learning, ggplot2, seaborn, matplotlib, Jupyter, RStudio.

Data Management:
SQL/NoSQL databases, data curation, QC workflows, reproducible documentation.

Soft Skills:
Clear communication, teamwork, technical writing, project management, problem-solving, adaptability.

Languages:
English (Full professional proficiency), Spanish (Native)

Selected Publications (14 total)

Gallardo-Becerra L, Cornejo-Granados F, Bikel S, Arenas I, López-Leal G, Alvarado-Gonzalez C, et al. Bioactive plasmid- and phage-encoded antimicrobial peptides (AMPs) in the human gut: a metatranscriptome–virome profiling reveals exploratory links to metabolic human diseases. Microb Ecol. 2025 Nov 28.

Gallardo-Becerra L, Cornejo-Granados F, Romero-Hidalgo S, Lopez-Zavala AA, Cota-Huízar A, Cervantes-Echeverría M, et al. Host genome drives the microbiota enrichment of beneficial microbes in shrimp: exploring the hologenome perspective. Anim Microbiome. 2025 May 22;7(1):50.

Gallardo-Becerra L, Cervantes-Echeverría M, Cornejo-Granados F, Vazquez-Morado LE, Ochoa-Leyva A. Perspectives in searching antimicrobial peptides (AMPs) produced by the microbiota. Microb Ecol. 2023 Dec 1;87(1):8.

Gallardo-Becerra L, Cornejo-Granados F, García-López R, Valdez-Lara A, Bikel S, Canizales-Quinteros S, et al. Metatranscriptomic analysis to define the Secrebiome, and 16S rRNA profiling of the gut microbiome in obesity and metabolic syndrome of Mexican children. Microb Cell Fact. 2020 Dec;19(1):61.

Gallardo-Becerra L, Cornejo-Granados F, Leonardo-Reza M, Ochoa-Romo JP, Ochoa-Leyva A. A meta-analysis reveals the environmental and host factors shaping the structure and function of the shrimp microbiota. PeerJ. 2018 Aug 10;6:e5382.